

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2011 question paper
for the guidance of teachers

9702 PHYSICS

9702/41

Paper 4 (A2 Structured Questions), maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- Cambridge will not enter into discussions or correspondence in connection with these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2011 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	41

Section A

- 1 (a) (i) force proportional to product of masses B1
force inversely proportional to square of separation B1 [2]
- (ii) separation much greater than radius / diameter of Sun / planet B1 [1]
- (b) (i) e.g. force or field strength $\propto 1/r^2$
potential $\propto 1/r$ B1 [1]
- (ii) e.g. gravitational force (always) attractive B1
electric force attractive or repulsive B1 [2]
- 2 (a) number of atoms of carbon-12 M1
in 0.012 kg of carbon-12 A1 [2]
- (b) $pV = NkT$ or $pV = nRT$ C1
substitutes temperature as 298 K C1
either $1.1 \times 10^5 \times 6.5 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times 298$
or $1.1 \times 10^5 \times 6.5 \times 10^{-2} = n \times 8.31 \times 298$ and $n = N / 6.02 \times 10^{23}$ C1
 $N = 1.7 \times 10^{24}$ A1 [4]
- 3 (a) acceleration / force proportional to displacement from a fixed point M1
acceleration / force (always) directed towards that fixed point / in opposite direction to displacement A1 [2]
- (b) (i) $A\rho g / m$ is a constant and so acceleration proportional to x B1
negative sign shows acceleration towards a fixed point / in opposite direction to displacement B1 [2]
- (ii) $\omega^2 = (A\rho g / m)$ C1
 $\omega = 2\pi f$ C1
 $(2 \times \pi \times 1.5)^2 = (\{4.5 \times 10^{-4} \times 1.0 \times 10^3 \times 9.81\} / m)$ C1
 $m = 50 \text{ g}$ A1 [4]
- 4 (a) work done in bringing unit positive charge M1
from infinity (to that point) A1 [2]
- (b) (i) field strength is potential gradient B1 [1]
- (ii) field strength proportional to force (on particle Q) B1
potential gradient proportional to gradient of (potential energy) graph B1
so force is proportional to the gradient of the graph A0 [2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	41

	(c) energy = $5.1 \times 1.6 \times 10^{-19}$ (J)	C1	
	potential energy = $Q_1 Q_2 / 4\pi\epsilon_0 r$	C1	
	$5.1 \times 1.6 \times 10^{-19} = (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times r$	C1	
	$r = 2.8 \times 10^{-10}$ m	A1	[4]
	(d) (i) work is got out as x decreases so opposite sign	M1 A1	[2]
	(ii) energy would be doubled gradient would be increased	B1 B1	[2]
5	(a) region (of space) where there is a force <i>either</i> on / produced by magnetic pole <i>or</i> on / produced by current carrying conductor / moving charge	M1 A1	[2]
	(b) (i) force on particle is (always) normal to velocity / direction of travel speed of particle is constant	B1 B1	[2]
	(ii) magnetic force provides the centripetal force $mv^2 / r = Bqv$ $r = mv / Bq$	B1 M1 A0	[2]
	(c) (i) direction from 'bottom to top' of diagram	B1	[1]
	(ii) radius proportional to momentum ratio = $5.7 / 7.4$ = 0.77 <i>(answer must be consistent with direction given in (c)(i))</i>	C1 A1	[2]
6	(a) (i) to concentrate the (magnetic) flux / reduce flux losses	B1	[1]
	(ii) changing flux (in core) induces current in core currents in core give rise to a heating effect	M1 A1	[2]
	(b) (i) e.m.f. induced proportional to rate of change of (magnetic) flux (linkage)	M1 A1	[2]
	(ii) magnetic flux in phase with / proportional to e.m.f. / current in primary coil e.m.f. / p.d. across secondary proportional to rate of change of flux so e.m.f. of supply not in phase with p.d. across secondary	M1 M1 A0	[2]
	(c) (i) for same power (transmission), high voltage with low current with low current, less energy losses in transmission cables	B1 B1	[2]
	(ii) voltage is easily / efficiently changed	B1	[1]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	41

- 7 (a) for a wave, electron can 'collect' energy continuously B1
for a wave, electron will always be emitted /
electron will be emitted at all frequencies..... M1
after a sufficiently long delay A1 [3]
- (b) (i) *either* wavelength is longer than threshold wavelength
or frequency is below the threshold frequency
or photon energy is less than work function B1 [1]
- (ii) $hc / \lambda = \phi + E_{\text{MAX}}$ C1
 $(6.63 \times 10^{-34} \times 3.0 \times 10^8) / (240 \times 10^{-9}) = \phi + 4.44 \times 10^{-19}$ C1
 $\phi = 3.8 \times 10^{-19} \text{ J (allow } 3.9 \times 10^{-19} \text{ J)}$ A1 [3]
- (c) (i) photon energy larger M1
so (maximum) kinetic energy is larger A1 [2]
- (ii) fewer photons (per unit time) M1
so (maximum) current is smaller A1 [2]
- 8 (a) (i) Fe shown near peak A1 [1]
- (ii) Zr shown about half-way along plateau A1 [1]
- (iii) H shown at less than 0.4 of maximum height A1 [1]
- (b) (i) heavy / large nucleus breaks up / splits M1
into two nuclei / fragments of approximately equal mass A1 [2]
- (ii) binding energy of nucleus = $B_E \times A$ B1
binding energy of parent nucleus is less than sum of binding energies
of fragments B1 [2]

Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	41

Section B

- 9 (a) to compare two potentials / voltages
output depends upon which is greater M1 A1 [2]
- (b) (i) resistance of thermistor = $2.5\text{ k}\Omega$
resistance of X = $2.5\text{ k}\Omega$ C1 A1 [2]
- (ii) at 5°C / at $< 10^\circ\text{C}$, $V^- > V^+$
so V_{OUT} is -9 V M1 A1
at 20°C / at $> 10^\circ\text{C}$, $V^- < V^+$ and V_{OUT} is $+9\text{ V}$ B1
 V_{OUT} switches between negative and positive at 10°C B1 [4]
(allow similar scheme if 20°C treated first)
- 10 (a) product of density (of medium) and speed of sound (in the medium) B1 [1]
- (b) α would be nearly equal to 1 M1
either reflected intensity would be nearly equal to incident intensity
or coefficient for transmitted intensity = $(1 - \alpha)$ M1
transmitted intensity would be small A1 [3]
- (c) (i) $\alpha = (1.7 - 1.3)^2 / (1.7 + 1.3)^2$
 $= 0.018$ C1 A1 [2]
- (ii) attenuation in fat = $\exp(-48 \times 2x \times 10^{-2})$ C1
 $0.012 = 0.018 \exp(-48 \times 2x \times 10^{-2})$ C1
 $x = 0.42\text{ cm}$ A1 [3]
- 11 (a) frequency of carrier wave varies M1
(in synchrony) with the displacement of the information signal A1 [2]
- (b) (i) 5.0 V A1 [1]
(ii) 640 kHz A1 [1]
(iii) 560 kHz A1 [1]
(iv) 7000 (condone unit) A1 [1]
- 12 (a) e.g. acts as 'return' for the signal
shields inner core from noise / interference / cross-talk
(any two sensible answers, 1 each, max 2) B2 [2]
- (b) e.g. greater bandwidth
less attenuation (per unit length)
less noise / interference
(any two sensible answers, 1 each, max 2) B2 [2]
- (c) attenuation is 2.4 dB C1
attenuation = $10\lg(P_1/P_2)$ C1
ratio = 1.7 A1 [3]